

Bright source simulations

Thomas Dauser
Remeis Observatory & ECAP

SIXTE Workshop — May 2026

Outline

We've prepared several simulation scripts which can be run and modified.

They are broadly categorized into

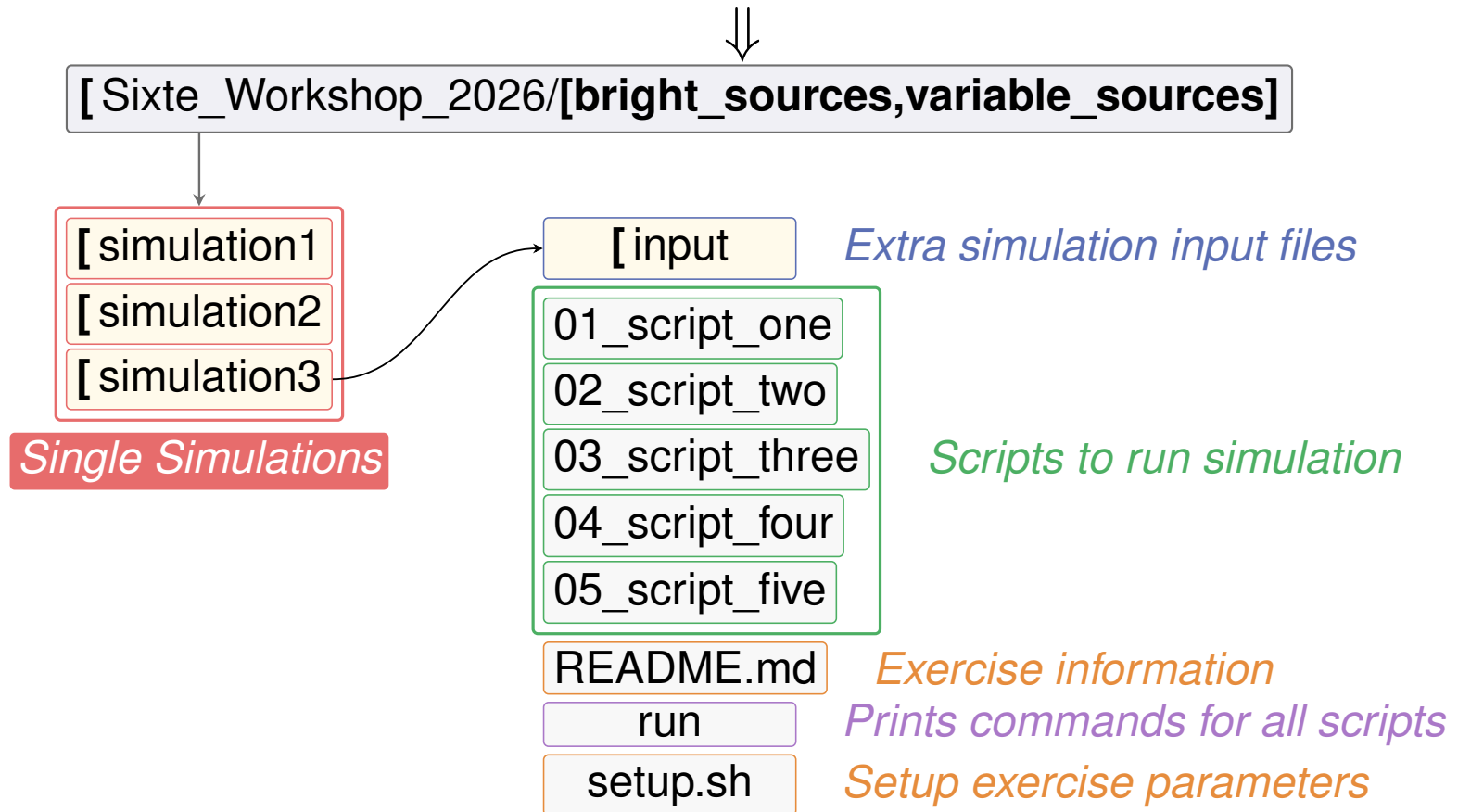
Bright Sources Examples that analyze important detector effects for the WFI (pile-up) and the X-IFU (branching ratios)

Variable Sources Creating and analyzing variable sources (light curves, PSDs, ...)

Detailed help and tutorials on all Sixte Tools are in the **Simulator Manual** (<https://www.sternwarte.uni-erlangen.de/sixte/sixte-manual/>)

Where to find the files: GitHub Repository

```
git clone https://github.com/stierhofj/Sixte_Workshop_2026.git
```



Simulation Examples

Bright Sources

[simple_src_xifu](#) basic simulations to understand grading and the branching ratios

[microcalorimeter](#) building upon the simple example, also extract spectra for different grades

[pileup](#) basic simulations to understand pileup with the WFI

[python_exercise](#) longer tutorial in exercise style to show how python can be used to create a SIMPUT for multiple sources, and analyze pile-up

Variable Sources

[periodic_lc](#) simple example that shows how to build a simput for a source with a periodic light curve

[psd](#) simple example that shows how to build a simput give a PSD

Bright Sources with *NewAthena*

(see the *WFI* and *X-IFU* introduction talks for the flux limits)

Important to choose the right *observation mode*: **defocused optics**

WFI: `fd_wfi_ff_df35mm.xml`

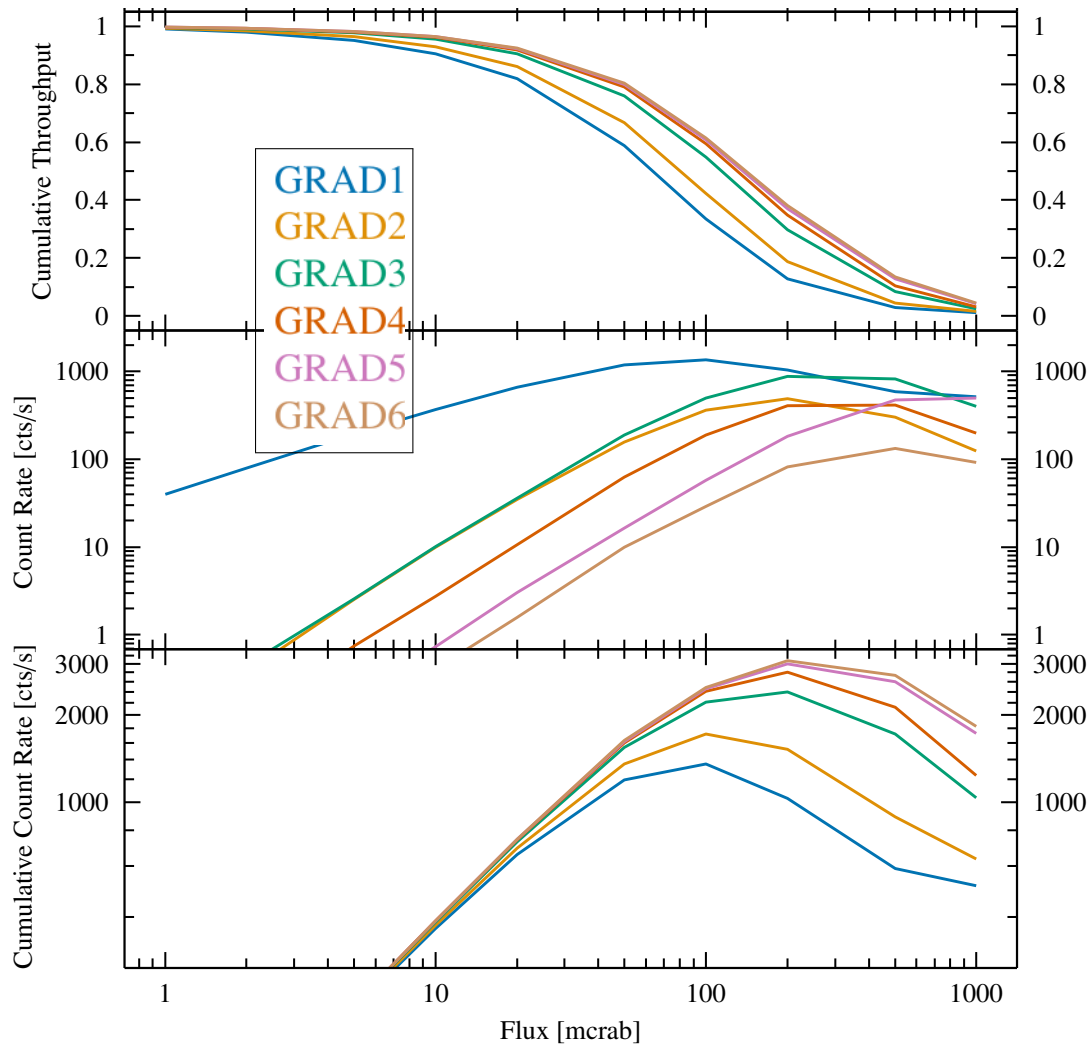
X-IFU: `xifu_optical_filter_defoc.xml`

For the brightest sources: use thick filter (removes all photons below 2 keV)

WFI: `fd_wfi_ff_df35mm_thick_filter.xml`

X-IFU: `xifu_be_filter_defoc.xml`

Observing Mode for X-IFU: Defocused incl. Optical Filter



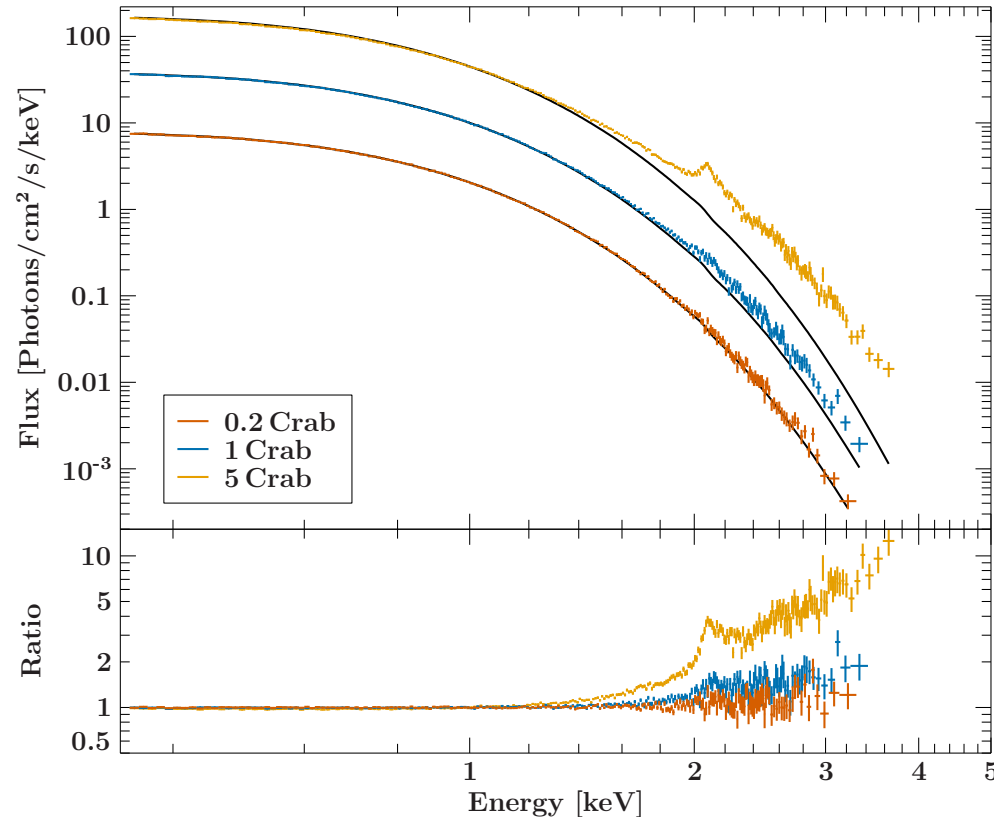
High resolution 4 eV
(GRAD1 & GRAD2) events
strongly suppressed
> 50 mCrab

⇒ use conservative approach to extract everything as **Intermediate Resolution 4.2 eV** (GRAD3) events

Cumulative Throughput/Counts
means for a given flux, throughput/Counts are added for the given GRADE X and better (meaning $\text{GRADE} \leq \text{GRADE } X$)

for bright 50-200 mCrab sources, intermediate resolution (4.2 eV) is ideal

The WFI and pile-up



pile-up distorts the spectral shape

the limits for the WFI are
(see presentation)

defocus XYZ

defocus + Be filter XYZ

ways to check if pile-up is a problem in simulations

1. verify that pile-up fraction is below 1% (see WFI presentation and Dauser et al., 2019 for more details), and exercise `pileup`
2. test that the simulated spectrum matches the input spectral shape